

MEDICAL STORE SAAS PLATFORM

Comprehensive Technical Documentation & System Implementation Manual (Phases 1 — 11)

Stack: Laravel 12 + MySQL + Blade (Vanilla CSS)	Completed Phases: 11 / 11 Modules (Super Admin)
Automated Test Suite: 143 Tests Passed (622 Assertions)	Architecture: Multi-Tenant SaaS with Strict RBAC
Project Directory: d:\projects\Medical Store	Status: Production Ready Platform Layer

1. System Architecture & Role Boundaries

The software is engineered as a robust, scalable Multi-Tenant SaaS system designed specifically for pharmacy retail and wholesale management. The core architectural boundary enforces complete separation between the **Platform Management Layer** (Super Admin) and the **Tenant Operations Layer** (Store Owners and Pharmacies).

Role / Entity	Domain Scope	Key Responsibilities & Boundary Enforcement
Super Admin /super-admin/*	Platform-wide (Cross-Tenant)	- Manages stores, owners, subscription tiers, payments & system health. - Strictly prohibited from directly creating store sales, medicines, or customer invoices. - Does NOT belong to any single store.
Store Owner /store/*	Single Tenant (Store Isolated)	- Full management of own pharmacy (Medicines, Suppliers, Batches, POS Sales). - Zero access to other stores' data; strictly scoped via store_id foreign key. - Can operate solo without hiring staff.
Store Staff (Optional)	Tenant Sub-User (Role Scoped)	- Pharmacists, Cashiers, Inventory clerks under a specific Store Owner. - Optional feature; system works 100% seamlessly for solo owners without staff creation.
Customer (Optional)	Store Client (Ledger Only)	- Optional profile created only when credit (Khata) or health records are tracked. - Walk-in customer POS sales work instantly without mandatory registration.

2. Database Schema & Relational Architecture

The relational database schema is normalized and indexed to guarantee sub-millisecond query latency and absolute data integrity across all tenant interactions.

Table Name	Primary Columns & Data Types	Foreign Keys & Constraints
users	id, name, email, mobile, password, role, status, store_id, remember_token, created_at	role IN ('super_admin','store_owner','staff') store_id -> stores.id (nullable)
stores	id, store_code (MED-XXXXXX), name, email, mobile, alt_mobile, address, city, state, pincode, gstin, drug_license_no, license_expiry_date, store_type, status, logo_path	UNIQUE(store_code) status IN ('active','inactive','suspended') INDEX(city, status)
subscription_plans	id, name, slug, description, price, billing_cycle, trial_days, max_staff, max_products, max_invoices, max_customers, features (JSON), status, is_popular	UNIQUE(slug) billing_cycle IN ('monthly','yearly','lifetime')

store_subscriptions	id, store_id, plan_id, start_date, end_date, trial_ends_at, status, notes	store_id -> stores.id plan_id -> subscription_plans.id status IN ('trial','active','expired','cancelled','suspended')
payments	id, store_id, subscription_id, plan_id, invoice_number, amount, currency, payment_method, transaction_id, payment_date, status, notes	store_id -> stores.id subscription_id -> store_subscriptions.id payment_method IN ('upi','card','net_banking','cash','bank_transfer')
notifications	id, type, priority, target, store_id, title, message, status, scheduled_at, sent_at, expires_at	target IN ('all','specific') priority IN ('normal','important','urgent') store_id -> stores.id (nullable)
settings	id, key, value, group, type, description	UNIQUE(key) group IN ('general','branding','contact','system','security','subscription')
audit_logs	id, user_id, action, module, description, ip_address, user_agent, old_values (JSON), new_values (JSON), created_at	user_id -> users.id (nullable on delete) INDEX(module, created_at, user_id) Immutable (No edit/delete)

3. Detailed Implementation Breakdown (Phases 1 to 11)

Below is the complete engineering summary of every phase implemented in the Super Admin platform:

Phase 1: Super Admin Authentication & Security Guard

Primary Controller: `app/Http/Controllers/SuperAdmin/Auth/LoginController.php`

- **Dedicated Auth Guard:** Configured `/super-admin/login` isolated from store owner login routes.
- **Public Registration Block:** Super Admin accounts cannot be self-registered; strictly seeded/provisioned.
- **Session & Middleware Security:** Protected via `SuperAdminMiddleware` ensuring store owners cannot escalate privileges.

Phase 2: Super Admin Dashboard & Live Telemetry

Primary Controller: `app/Http/Controllers/SuperAdmin/DashboardController.php`

- **Live Aggregation:** Computes real-time KPIs (Total/Active/Inactive Stores, Active Subscriptions, Expiring Soon, Total Platform Revenue).
- **Recent Activity Stream:** Real-time audit log feed rendering recent store creation, payments, and plan changes.
- **UI Theme:** Pure Fullscreen responsive layout strictly styled in Navy Blue & White palette.

Phase 3: Medical Store Management

Primary Controller: `app/Http/Controllers/SuperAdmin/StoreController.php`

- **Automatic Code Generator:** Creates unique sequential store codes like MED-000001, MED-000002.
- **Regulatory Compliance Fields:** GSTIN, Drug License Number, License Expiry Date, and Store Type (Retail/Wholesale/Both).
- **Branding & Status Controls:** Store logo file uploads, status switches (Active, Inactive, Suspended), city/state filters, and pagination.

Phase 4: Store Owner Management & Tenant Linking

Primary Controller: `app/Http/Controllers/SuperAdmin/StoreOwnerController.php`

- **Tenant Association:** Links a user account directly to a specific `store_id`.
- **Secure Password Hashing:** Bcrypt hashing with confirmation validation and status toggling.
- **Isolation Verification:** Ensures a Store Owner only sees their own store upon logging in to `/store/login`.

Phase 5: Subscription Plan Management Engine

Primary Controller: `app/Http/Controllers/SuperAdmin/SubscriptionPlanController.php`

- **Flexible Tier Architecture:** Create plans (Basic, Professional, Enterprise, Custom) with customizable billing cycles (Monthly, Yearly).
- **Quota Enforcement:** Configurable caps on Max Staff, Max Products (Medicines), Max Invoices/Sales, and Max Customers.
- **JSON Feature Flags:** Dynamic toggling of modules (Inventory, Purchase, Barcode, GST Reports, Staff Management).

Phase 6: Store Subscription Assignment & Lifecycle

Primary Controller: `app/Http/Controllers/SuperAdmin/StoreSubscriptionController.php`

- **Lifecycle Management:** Assign plans to stores with automated start/end dates, trial periods, and status tracking (Trial, Active, Expired, Suspended).
- **Overlap & Conflict Prevention:** Handles existing active subscriptions cleanly and maintains full historical logs of renewals.
- **Expiring Subscriptions Detection:** Instant detection of stores reaching expiry within 7 to 15 days.

Phase 7: Payment & Revenue Management

Primary Controller: `app/Http/Controllers/SuperAdmin/PaymentController.php`

- **Multi-Channel Payment Recording:** Records payments via UPI, Credit/Debit Card, Net Banking, Cash, and Bank Transfers.
- **Receipt & Invoice Generation:** Auto-generated transaction IDs and linked subscription IDs.
- **Payment Status Tracking:** Paid, Pending, Failed, Refunded, Cancelled with date and store filtering.

Phase 8: Reports, Analytics & CSV Export

Primary Controller: `app/Http/Controllers/SuperAdmin/ReportController.php`

- **Multi-Dimensional Analytics:** Overall platform health, Geographic Store Reports (City/State wise), Plan Share breakdown.
- **Financial Revenue Analytics:** Monthly revenue trends, payment method share, and date range filters.
- **Export Capability:** Direct streaming CSV export for Store, Subscription, and Payment reports.

Phase 9: Platform Notification Management

Primary Controller: `app/Http/Controllers/SuperAdmin/NotificationController.php`

- **Broadcast vs. Targeted Messaging:** Send notifications to All Stores or target a single specific Store.
- **Priority & Type Classification:** Urgent, Important, Normal priority with categories (Subscription Expiry, System Maintenance, Updates).
- **Lifecycle Status:** Draft, Scheduled, Sent, Cancelled with expiry scheduling.

Phase 10: Global System Settings & Maintenance

Primary Controller: `app/Http/Controllers/SuperAdmin/SettingController.php`

- **Categorized Configuration:** General (App name, Currency, Timezone), Branding (Logo, Favicon uploads), Contact Support, Security.
- **Maintenance Mode Switch:** Global toggle to put public/store tenant access in maintenance without locking out Super Admin.
- **SaaS Policies:** Default trial days, subscription expiry warning threshold, and registration toggles.

Phase 11: Enterprise Audit Logs & Sensitive Redaction

Primary Controller: `app/Http/Controllers/SuperAdmin/AuditLogController.php`

- **Immutable Security Trace:** Automatic logging of all Super Admin write/update/delete operations with IP address & User-Agent.
- **Field-Level Diffs:** Stores old_values and new_values in JSON format with interactive visual comparison modal.
- **Zero Leakage Policy:** Strictly redacts sensitive credentials (passwords, tokens, keys) before writing to audit database.

4. Real-World Onboarding Workflow (Walkthrough)

To understand how the 11 Super Admin modules interact in production, here is the complete end-to-end execution flow when onboarding a new medical store (**Sharma Medical Store, Varanasi**):

Step #	Action & Screen	Data Input & System Execution	Resulting System State
Step 1	Create Store /super-admin/stores/create	- Name: <i>Sharma Medical Store</i> - City: <i>Varanasi</i>, State: <i>UP</i> - Drug License: <i>DL-2026-9871</i> - GSTIN: <i>09ABCDE1234F1Z5</i>	System generates MED-000001. Store saved with status Active . Audit log recorded.
Step 2	Create Store Owner /super-admin/store-owners/create	- Name: <i>Rahul Sharma</i> - Email: <i>rahul@gmail.com</i> - Linked Store: <i>Sharma Medical Store</i> - Password: <i>*****</i>	Owner created with role store_owner. Tied to store_id = 1. Audit log recorded.
Step 3	Subscription Plan /super-admin/subscription-plans	Super Admin selects existing Professional plan (₹999/mo, 5 staff, 5000 medicines, unlimited billing).	Plan rules & quotas ready for assignment.
Step 4	Assign Subscription /super-admin/subscriptions/create	- Store: <i>Sharma Medical Store</i> - Plan: <i>Professional Plan</i> - Period: <i>01-09-2026 to 01-10-2026</i> - Status: <i>Active</i>	Store subscription activated. Expiry tracked for renewal alerts.
Step 5	Record Payment /super-admin/payments/create	- Amount: ₹999 - Method: <i>UPI</i> (TXN987654321) - Linked to Subscription ID #1 - Status: <i>Paid</i>	Revenue updated on Dashboard. Invoice marked as Paid.
Step 6	Store Login & POS /store/login	Rahul Sharma logs in with his credentials. Software loads isolated Store Owner Dashboard.	Rahul manages medicine stock, sales, and purchases for Sharma Medical Store exclusively.

5. Super Admin Complete Route Map

All Super Admin endpoints are prefixed with /super-admin and protected by authentication & role middlewares.

Route URI	HTTP Method	Controller Action	Description / Purpose
/super-admin/login	GET / POST	LoginController	Super Admin Auth & Session Start
/super-admin/dashboard	GET	DashboardController@index	Live Metrics & Activity Feed
/super-admin/stores/*	RESOURCE	StoreController	CRUD, Status Toggle, Auto Code
/super-admin/store-owners/*	RESOURCE	StoreOwnerController	Owner Accounts & Tenant Linking
/super-admin/subscription-plans/*	RESOURCE	SubscriptionPlanController	Plan Tiers, Quotas & Features
/super-admin/subscriptions/*	RESOURCE	StoreSubscriptionController	Assignment, Renewal, Status
/super-admin/payments/*	RESOURCE	PaymentController	Payment Records, Invoicing
/super-admin/reports/*	GET	ReportController	Store/Plan/Revenue Analytics & CSV
/super-admin/notifications/*	RESOURCE	NotificationController	Broadcast & Targeted Notifications
/super-admin/settings	GET / POST	SettingController	Global Config, Branding, Maintenance
/super-admin/audit-logs/*	GET	AuditLogController	Activity Journal & Diff Inspector

6. Quality Assurance & Automated Verification Suite

The codebase has been verified with comprehensive unit and feature test suites covering authentication, validation rules, authorization guards, lifecycle state machines, exports, and security redactions.

Total Automated Tests: 143 Tests			Total Assertions: 622 Assertions			Success Rate: 100% Passing		
Module / Component Test Suite			Tests			Key Validated Test Cases		
Super Admin Authentication			12			Login validation, guard isolation, invalid credentials, redirect flow, logout.		
Super Admin Dashboard			8			KPI calculations from real DB rows, revenue aggregations, recent audit feed.		
Store Management			18			Sequential code generation, unique constraints, GSTIN & Drug license validation, search/filters.		
Store Owner Management			14			Tenant linking, unique email check, password hashing, role enforcement.		
Subscription Plan Engine			15			Slug generation, quota validation, feature toggles, popular tag toggles.		
Store Subscription Assignment			16			Date calculation, trial expiration, status state machines, history tracking.		
Payment Management			15			Payment recording, transaction ID uniqueness, revenue link to subscription.		
Reports & Analytics			13			Multi-filter reporting, date ranges, CSV export headers and data integrity.		
Notification Engine			12			Target resolution (all vs specific), priority categorization, scheduled vs sent states.		
System Settings			10			Key-value persistence, branding file upload validation, maintenance mode toggle.		
Audit Logs & Redaction			10			Immutable records, telemetry capture, sensitive field redaction, JSON diff format.		

7. Future Development Roadmap (Store Owner Layer)

With the 11 foundational Super Admin modules fully operational, the system is perfectly poised for the development of the **Store Owner & Pharmacy Operations Layer**:

- **Phase 12 — Medicine & Drug Inventory Management:** Brand names, generic salt compositions, batch numbers, expiry dates, rack/shelf locations, and low-stock alerts.
- **Phase 13 — Supplier & Purchase Orders (B2B):** Wholesale vendor directories, purchase invoices, batch stock entry, and accounts payable.
- **Phase 14 — Point of Sale (POS) & Rapid Billing:** Barcode scanning, thermal receipt printing, GST calculation (CGST/SGST/IGST), and walk-in customer sales.
- **Phase 15 — Customer Accounts & Khata (Credit Ledger):** Patient prescription logs, credit balance tracking, and WhatsApp invoice sharing.
- **Phase 16 — Store Level Analytics & Tax Filing:** GSTR-1 & GSTR-3B tax export reports, daily sales summaries, and fast-moving medicine reports.

CORE ARCHITECTURAL RULE:

Super Admin = Platform Owner (Manages stores, licenses, plans, subscriptions, and platform stability).
Store Owner = Pharmacy Owner (Manages medicine inventory, purchases, sales, and billing in total isolation).
Solo Pharmacists operate with zero mandatory staff or customer overhead.